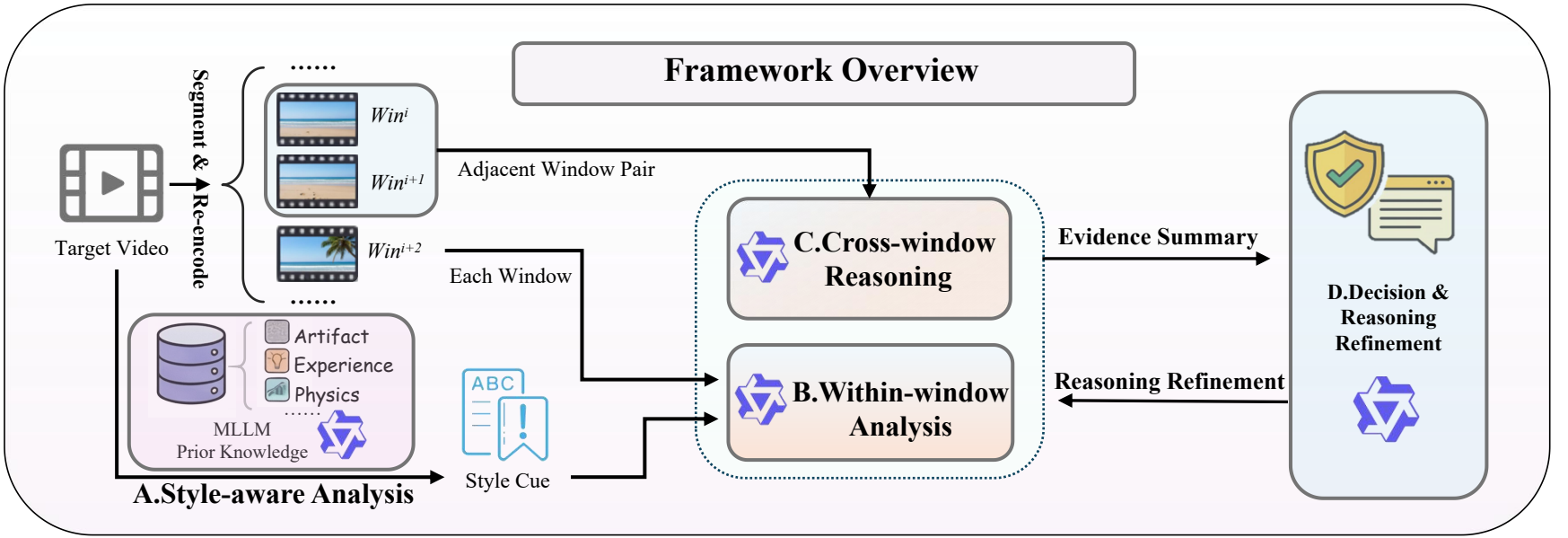
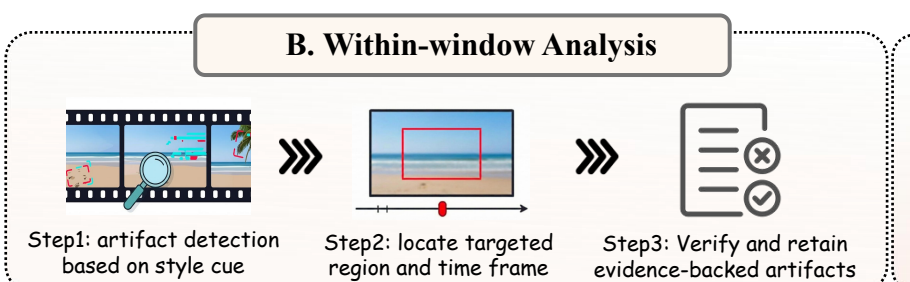
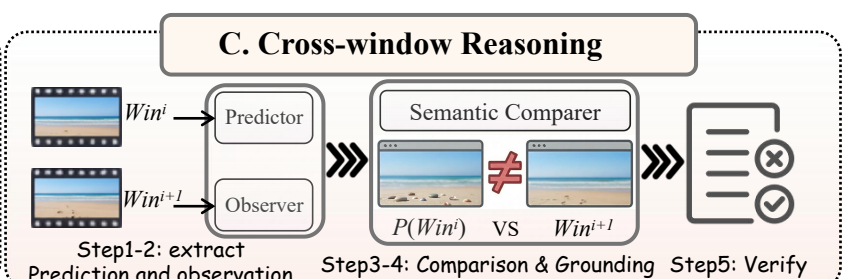

Framework Overview
Target Video
Segment & Re-encode
Win^i
Win^{i+1}
Win^{i+2}
Adjacent Window Pair
Each Window
Artifact
Experience
Physics
MLLM
Prior Knowledge
A.Style-aware Analysis
ABC
Style Cue
C.Cross-window Reasoning
B.Within-window Analysis
Evidence Summary
Reasoning Refinement
D.Decision & Reasoning Refinement


B. Within-window Analysis
Step1: artifact detection based on style cue
Step2: locate targeted region and time frame
Step3: Verify and retain evidence-backed artifacts


C. Cross-window Reasoning
Win^i
Win^{i+1}
Predictor
Observer
Step1-2: extract Prediction and observation
Semantic Comparer
P(Win^i) VS Win^{i+1}
Step3-4: Comparison & Grounding
Step5: Verify